Artificial Light Exposure During Pregnancy and Infant Health*

Andrés Azqueta-Gavaldón†
Banco de España

Ella Reese-Clauson‡
Colorado College

Guanyi Yang§
Colorado College

March 4, 2026 Preliminary Draft
Updated Version Here

Abstract

Artificial light at night (ALAN) is a pervasive feature of modern environments, yet its effects on fetal development remain poorly understood. Using U.S. birth records linked to satellite-based measures of gestational nighttime radiance, we estimate the impact of prenatal light exposure on infant health. A 10 percent increase in average gestational radiance reduces birth weight by roughly 0.4 to 2 percent (13 to 65 grams at the mean) and increases neonatal intensive care unit admission by about 1 to 3 percentage points. Effects are largest in urban and high air-pollution areas and among infants born to non-Hispanic Black mothers. We estimate these effects using flexible machine-learning methods and complement them with an instrumental-variables strategy that exploits quasi-random variation in the satellite retrieval process to assess attenuation from measurement error.

Keywords: Artificial light at night, light pollution, infant health, prenatal environmental exposure, birth outcomes, environmental inequality.

JEL Codes: I10, Q53, J13, I14.

* ***Acknowledgments:*** We thank the State of the Rockies and the Hulbert Center for Southwest Studies at Colorado College for their support. ***Disclaimer:*** The views expressed in this paper are those of the authors and do not necessarily reflect the positions of the Central Bank of Spain. All errors are our own.

†Banco de España, C. de Alcalá, 522, 28027 Madrid, Spain. Email: andres.azqueta@bde.com.

‡Colorado College, Palmer Hall 101-A 14 E. Cache la Poudre St. Colorado Springs, CO 80903, USA. Email: e_reeseclauson@coloradocollege.edu.

§Colorado College, Palmer Hall 114-C 14 E. Cache la Poudre St. Colorado Springs, CO 80903, USA. Email: gyang@ColoradoCollege.edu.

1 Introduction

Artificial light at night has become a defining feature of modern economic life. Urbanization, electrification, and nighttime economic activity have dramatically increased ambient light exposure, reshaping the environments in which people live, work, and sleep. While a growing literature links artificial light to adult outcomes such as sleep disruption and cancer risk,[1] far less is known about its consequences for fetal development. This omission is striking given that pregnancy is a period of heightened biological sensitivity and that exposure to nighttime light is nearly universal in developed economies.

This paper studies whether prenatal exposure to artificial light at night affects infant health. Focusing on birth outcomes that are strongly predictive of later-life health and human capital—birth weight, gestational length, prematurity, and neonatal intensive care unit (NICU) admission[2]—we show that higher gestational light exposure worsens infant health. Using national U.S. birth records linked to satellite-based measures of nighttime illumination during gestation, we document economically meaningful effects concentrated in high-exposure environments.

Estimating the impact of light exposure poses two fundamental challenges. First, nighttime brightness is tightly linked to economic development. Brighter areas tend to be wealthier and better resourced, which improves infant health, but they are also more urban, polluted, and stressful, which worsens health. The net relationship between light and birth outcomes is therefore ambiguous *a priori* and likely nonlinear. Second, satellite measures of nighttime radiance are noisy. Cloud cover, lunar illumination, and atmospheric conditions distort observed brightness, creating measurement error that can substantially attenuate estimated effects. These challenges have limited existing evidence to small-scale or observational studies that are difficult to generalize.

We address these issues using population-scale data and complementary empirical approaches. Our primary analysis uses the universe of U.S. births from the National Vital Statistics System between 2014 and 2018, over 12 million observations, linked to high-resolution monthly satellite measures of nighttime radiance. To flexibly account for the complex correlation between light exposure, socioeconomic conditions, and environmental factors, we estimate effects using double machine learning (DML), which allows us to control for a rich set of maternal, demographic, economic, and environmental covariates without imposing restrictive functional forms. This approach provides stable population-average estimates and enables detailed heterogeneity analysis.

To address measurement error in satellite-based light exposure, we complement

[1] Medical and biological research shows that exposure to artificial light at night can disrupt circadian rhythms and sleep patterns (Cho et al., 2015; Blume et al., 2019). These pathways are plausibly relevant during pregnancy, though our analysis does not isolate specific biological mechanisms.

[2] A large economics literature shows that in utero exposure to environmental stressors has persistent effects on infant health and later-life outcomes, including evidence from traffic congestion and air pollution (Currie and Walker, 2011; Deschênes et al., 2017).

this analysis with an instrumental variables strategy that exploits quasi-random variation in the satellite measurement process that changes the precision with which radiance is recorded. Because these fluctuations reflect observation conditions and retrieval quality rather than changes in underlying ground illumination, they generate plausibly exogenous variation in measured radiance that is orthogonal to local economic activity and baseline health determinants. This measurement-process variation provides a complementary source of identifying variation that is less exposed to slow-moving unobserved local factors correlated with both nighttime illumination and infant health. This design allows us to assess the extent of attenuation from measurement error and provides an independent validation of the main results. Across approaches, the evidence consistently points to substantial adverse effects of prenatal light exposure.

Across specifications, a 10 percent increase in average nighttime radiance during pregnancy reduces birth weight by roughly 0.4 to 2 percent and increases the probability of NICU admission by about 1 to 3 percentage points. Effects on gestational length, low birth weight, and preterm birth are smaller but consistently negative. Estimates are considerably larger when flexible adjustment and measurement error correction are applied, indicating that conventional linear models substantially understate the health costs of light exposure.

The impacts of prenatal light exposure are highly heterogeneous. Adverse effects are concentrated in urban areas and in counties with poor air quality, suggesting that light pollution interacts with other environmental stressors. We also find that infants born to non-Hispanic Black mothers experience substantially larger effects, highlighting light exposure as a previously underappreciated contributor to disparities in early-life health. Moreover, the relationship between light exposure and birth outcomes is nonlinear: harmful effects emerge primarily at higher exposure levels, consistent with threshold or cumulative-stress mechanisms.

This paper contributes to the literature on environmental determinants of health by identifying artificial light at night as an important and nearly universal prenatal exposure with significant consequences for infant health. It also contributes methodologically by demonstrating how modern machine learning tools and physically grounded measurement strategies that leverage the satellite retrieval process can be combined to study environmental exposures measured through remote sensing. The DML estimates identify population-average effects under a selection-on-observables assumption, while the instrumental-variables estimates identify a local average treatment effect driven by variation in satellite observation quality. Substantively, our results suggest that policies regulating nighttime illumination, often motivated by energy use or aesthetics, may have meaningful public health benefits, particularly in high-exposure and disadvantaged communities.

The closest study to ours is Argys et al. (2021), which examines light pollution and birth outcomes using a direct measure of skyglow and an instrumental-variables strategy based on Walker's law from physics. We complement this work by studying a distinct measure of nighttime illumination, satellite-based radiance from VIIRS, aligned to the gestational window in population-scale U.S. natality data. Methodolog-

ically, we combine flexible adjustment for high-dimensional confounding using double machine learning with an independent measurement-based strategy that exploits variation in satellite observation quality to assess attenuation from measurement error. Despite differences in measurement and identification, both papers reach the same qualitative conclusion: greater nighttime light exposure during pregnancy worsens infant health.

The remainder of the paper proceeds as follows. Section 2 describes the data and construction of gestational light exposure measures. Section 3 outlines the empirical strategy. Section 4 presents the main results and heterogeneity analyses. Section 5 reports robustness checks. Section 6 concludes.

2 Data

This section describes the birth records, satellite-based measures of nighttime light, and additional county-level covariates used in the analysis. Details on identification, measurement-error correction, and estimation are deferred to Section 3 and the Appendix.

2.1 Birth Records

Our primary data come from restricted-use natality microdata from the National Vital Statistics System (NVSS), which records the universe of live births in the United States. The NVSS includes detailed information on infant health outcomes, maternal and paternal characteristics, and clinical conditions recorded at delivery. Because reporting standards are uniform across states and over time, these data are widely used in research on infant health and the fetal origins of later-life outcomes.

We focus on births occurring between January 2014 and December 2018. This window is dictated by the availability of consistent satellite-based measures of nighttime radiance and observation-quality variables. After applying standard sample restrictions and merging to environmental and county-level covariates, the final analysis sample contains 12,027,573 births.

The main infant health outcomes studied are birth weight, gestational length, indicators for low birth weight and preterm birth, and admission to a neonatal intensive care unit (NICU). These outcomes are standard in the fetal origins literature and are strong predictors of later-life health and economic outcomes.

2.2 Satellite Measures of Nighttime Light

We measure ambient nighttime illumination using satellite radiance from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band, drawing on the Black Marble product suite developed by NASA (Román et al., 2018). Specifically, we use the monthly stray-light–corrected VIIRS composite (VCMSLCFG), which provides cloud-screened and radiometrically calibrated estimates of nighttime radiance at global scale. The stray-light correction removes seasonal light contamination at

high latitudes and improves comparability of radiance across months and years. We aggregate these data to the county–month level. Radiance is measured in nanowatts per square centimeter per steradian ($nW/cm^2/sr$), where higher values correspond to brighter nighttime environments. Details on radiometric calibration, quality flags, and cloud masking are provided in the Black Marble User Guides (Román and Team, 2019, 2021). Satellite-based nighttime lights have been widely used in economics to proxy for economic activity and development, reflecting their strong correlation with population density and income (Henderson et al., 2012; Chen and Nordhaus, 2019). In this paper, we use these data not as a proxy for economic activity, but as a direct measure of ambient nighttime illumination during pregnancy.

For each birth, we construct *average gestational radiance* by linking the mother's county of residence to county–month radiance values for the months of pregnancy and averaging across observed gestational months. Pregnancy months are defined using gestational age and the infant's month of birth. For preterm births, months prior to conception are treated as missing so that exposure measures reflect only plausible in utero periods (see Appendix A for details).

Satellite-based nighttime radiance captures outdoor artificial illumination emitted upward or reflected from the earth's surface, including street lighting, commercial and industrial activity, and other sources of ambient nighttime brightness. These measures do not capture indoor light exposure directly and cannot distinguish between individual-level behaviors or housing characteristics that mediate exposure, such as window shielding or bedroom lighting. Because the NVSS reports maternal residence at delivery rather than complete residential histories, exposure is measured at the county–month level, which is the finest spatial resolution that can be consistently linked to the universe of U.S. births. This aggregation introduces several sources of measurement error, including within-county heterogeneity, residential mobility during pregnancy, and imperfect mapping between outdoor and indoor light environments. These forms of error are unlikely to be systematically correlated with infant health conditional on controls and are expected to attenuate estimated effects toward zero rather than generate spurious negative associations.

2.3 Observation Quality and Measurement Noise

Satellite-based radiance measures are subject to nontrivial measurement noise. Cloud cover, lunar illumination, atmospheric conditions, and satellite viewing geometry affect how reliably the earth's surface is observed, generating variation in recorded brightness that is unrelated to true ground-level light emissions.

The VIIRS Black Marble product includes *Mandatory Quality Flags* (MQF) that summarize observation conditions at the pixel level. Observation quality is derived from the VIIRS Black Marble daily product (VNP46A2), described in Appendix B. Using daily Black Marble data, we compute county–month shares of observations with high-quality retrievals (MQF$=0$) and lower-quality retrievals (MQF$=1$), and aggregate these measures over the gestational window. These measures capture fluctuations in measurement reliability rather than changes in underlying illumination, forming the basis for the instrumental-variables strategy described in Section 3.

2.4 County-Level Covariates

We merge the birth records to county-level economic and environmental variables. Economic covariates include the local unemployment rate, county population, and real per-capita personal income. Environmental conditions are captured using the Environmental Protection Agency's Air Quality Index (AQI). Because AQI is not observed for all counties and months, regressions include an indicator for missing AQI observations and set missing AQI values to zero for estimation. Descriptive statistics for AQI are computed using observed values only.

2.5 Descriptive Statistics

Table 1 reports descriptive statistics for all variables used in the main specifications. Birth outcomes exhibit standard magnitudes in national natality data. Mean birth weight is 3,311 grams and mean gestational length is 38.8 weeks. Prematurity occurs in 9.6% of births, low birth weight in 6.4%, and NICU admission in 7.4%.

Average radiance during gestation is 7.78 nW/cm^2/sr, with substantial cross-county variation. Mothers are 29.1 years old on average, with a pre-pregnancy BMI of 26.3 and approximately 11.5 prenatal visits. The county-level unemployment rate averages 4.6%, and county population and income vary widely, reflecting the national scope of the data. Average AQI is 51.8 in the subsample with observed values, and missingness is tracked separately. The MQF index in Panel B of Table 1 is constructed from Black Marble quality flags aggregated and standardized over the gestational window. Detailed construction is in Appendix B.

3 Empirical Strategy

Our objective is to estimate the causal effect of prenatal exposure to artificial light at night on infant health outcomes. Identification is challenging because nighttime illumination is spatially correlated with economic development, urban density, and other environmental exposures, and because satellite-based measures of radiance are subject to nonclassical measurement error. We therefore adopt a complementary empirical strategy that combines flexible selection-on-observables identification with an instrumental variables design. These approaches rely on different identifying assumptions and correspond to different causal estimands, and their comparison is informative about the sources of bias in conventional estimates.

3.1 Baseline Framework

Let Y_{icst} denote an infant health outcome for birth i occurring in county c, state s, and year–month t. Outcomes include birth weight, gestational length, indicators for low birth weight and preterm birth, and NICU admission. Let L_{icst} denote the logarithm of average nighttime radiance during the gestational window for the mother's county of residence. The baseline estimating equation is

$$Y_{icst} = \beta\, L_{icst} + X'_{icst}\gamma + \mu_s + \lambda_t + \varepsilon_{icst}, \tag{1}$$

Table 1: Descriptive Statistics

Variable	Mean	SD	N
Panel A. Infant Outcomes			
Birth weight (g)	3311.3	551.7	12,027,573
Gestation (weeks)	38.8	2.19	12,027,573
Low birth weight (< 2500 g)	0.064	0.245	12,027,573
Preterm (< 37 weeks)	0.096	0.295	12,027,573
NICU admission	0.074	0.262	12,027,573
Apgar score (5 min)	8.84	0.67	12,027,573
Panel B. Light Exposure and Instrument			
Gestational radiance (nW/cm^2/sr)	7.78	12.81	12,027,573
Dynamic MQF index (standardized)	-0.004	0.985	12,027,573
Panel C. Infant Characteristics			
Female infant	0.488	0.500	12,027,573
Cyanotic congenital heart condition	0.0002	0.022	12,027,573
Panel D. Maternal Characteristics			
Mother's age (years)	29.08	5.66	12,027,573
Pre-pregnancy BMI	26.30	6.52	12,027,573
Pregnancy weight gain (lbs)	30.00	14.71	12,027,573
Prenatal visits	11.51	3.94	12,027,573
Prior live births	1.09	1.25	12,027,573
Hispanic	0.233	0.423	12,027,573
Non-Hispanic Black	0.111	0.314	12,027,573
Non-Hispanic Asian	0.070	0.255	12,027,573
American Indian/Alaska Native	0.007	0.084	12,027,573
Native Hawaiian/Pacific Islander	0.002	0.047	12,027,573
Mixed race	0.020	0.140	12,027,573
Less than HS	0.116	0.321	12,027,573
College degree	0.312	0.463	12,027,573
Advanced degree	0.097	0.170	12,027,573
No reported risk factors	0.705	0.456	12,027,573
Panel E. Paternal Characteristics			
Father's age (years)	31.50	6.77	12,027,573
Panel F. Environmental and County Controls			
AQI (observed)	51.84	17.83	10,443,215
AQI missing indicator	0.132	0.338	12,027,573
Unemployment rate (%)	4.59	1.54	12,027,573
County population	1,293,094	2,014,237	12,027,573
Real per-capita income (2017$)	52,550	18,639	12,027,573

Notes: Observations span January 2014–December 2018. Radiance is averaged over the gestational window. The dynamic MQF index is standardized by construction. AQI statistics are computed on observed values only; regressions include an AQI missing indicator.

where X_{icst} includes maternal and paternal characteristics, race and education indicators, and county-level economic and environmental controls. State fixed effects μ_s absorb time-invariant differences in geography, health systems, and reporting practices, while year–month fixed effects λ_t capture national shocks, seasonality, and changes in satellite processing or clinical practice.

Ordinary least squares estimation of equation (1) is unlikely to recover a causal effect. First, the relationship between infant health and observed covariates is plausibly nonlinear and high dimensional, making linear adjustment sensitive to functional-form misspecification. Second, nighttime radiance is measured with error due to cloud cover, lunar illumination, and satellite viewing geometry, which can attenuate estimated effects. Finally, unobserved neighborhood characteristics or housing conditions may remain correlated with both illumination and infant health even after conditioning on rich controls.

3.2 Causal Effects Under Selection on Observables: Double Machine Learning

Our primary observational identification strategy estimates population-average causal effects under a selection-on-observables assumption: conditional on X_{icst}, residual variation in gestational radiance is assumed independent of potential outcomes. To implement this strategy while remaining robust to high-dimensional confounding and nonlinearities, we use Double Machine Learning (DML) as developed by Chernozhukov et al. (2018).

We adopt the partially linear model

$$Y_i = \theta\, L_i + g(X_i) + u_i, \tag{2}$$

where $g(\cdot)$ is an unknown function of observed covariates. DML combines flexible machine learning estimation of nuisance functions with orthogonalized estimating equations that are locally insensitive to regularization bias. This approach yields valid $\sqrt{n}$-consistent inference on θ even when $g(\cdot)$ is estimated using complex learners.

Let

$$m(X_i) = \mathbb{E}[L_i \mid X_i], \qquad \ell(X_i) = \mathbb{E}[Y_i \mid X_i]. \tag{3}$$

Using K-fold cross-fitting, we estimate $\widehat{m}(X_i)$ and $\widehat{\ell}(X_i)$ on auxiliary samples and construct residuals

$$\widetilde{L}_i = L_i - \widehat{m}(X_i), \qquad \widetilde{Y}_i = Y_i - \widehat{\ell}(X_i). \tag{4}$$

The DML estimator is

$$\widehat{\theta}_{\mathrm{DML}} = \frac{\sum_{i=1}^{n} \widetilde{L}_i\, \widetilde{Y}_i}{\sum_{i=1}^{n} \widetilde{L}_i^2}. \tag{5}$$

We implement DML with five-fold cross-fitting and estimate nuisance functions using Random Forests (Breiman, 2001), Gradient Boosting (Friedman, 2001), and Lasso (Tibshirani, 1996). Random Forests are our preferred specification due to their

stability and ability to capture nonlinear interactions. Results are highly similar across learners, indicating that findings are not driven by a particular modeling choice. Additional implementation details are provided in Appendix D.

3.3 Causal Effects Under an Exclusion Restriction: Instrumental Variables

Selection-on-observables may fail if unobserved factors remain correlated with both illumination and infant health, and measurement error in satellite radiance may attenuate both OLS and DML estimates. To address these concerns, we complement the DML analysis with an instrumental variables strategy that exploits exogenous variation in satellite observation quality.

Let Z_{icst} denote a dynamic, normalized index of satellite observation quality aggregated over the gestational window. This index is constructed from county–month shares of high-quality VIIRS Black Marble retrievals based on the Mandatory Quality Flag, residualized on state and year–month fixed effects and standardized prior to aggregation. Construction of Z_{icst} is described in Appendix B.

Conceptually, the MQF index is a shifter of measurement quality: it moves the observed radiance signal closer to (or farther from) true latent illumination without directly affecting local light production. Under this interpretation, the exclusion restriction is that observation quality affects infant outcomes only through its impact on measured radiance, not through independent changes in economic activity, health care access, or other determinants of infant health.

This IV differs from Argys et al. (2021), who instrument skyglow using Walker's law; our instrument instead leverages quasi-random variation in the satellite measurement process to isolate variation in *measured* radiance driven by the satellite retrieval process rather than by changes in underlying illumination.

We estimate the first stage

$$L_{icst} = \pi\, Z_{icst} + X'_{icst}\delta + \mu_s + \lambda_t + \eta_{icst}, \tag{6}$$

and the second stage

$$Y_{icst} = \beta^{IV}\, \widehat{L}_{icst} + X'_{icst}\gamma^{IV} + \mu_s + \lambda_t + \varepsilon^{IV}_{icst}, \tag{7}$$

where $\widehat{L}_{icst}$ is the fitted value from equation (6). Under this exclusion restriction, β^{IV} identifies a local average treatment effect for births whose measured exposure is shifted by variation in observation quality.

First-stage estimates and weak-identification diagnostics are reported in Appendix C. Appendix Table 6 reports the baseline first stage, and Appendix Table 7 reports results for alternative exposure definitions. These diagnostics indicate that the instrument is relevant and behaves as expected, though first-stage strength varies across specifications.

3.4 Interpretation Across Estimators

OLS, DML, and IV estimates correspond to distinct identifying assumptions and estimands. OLS relies on linear adjustment and is sensitive to functional-form misspecification and measurement error. DML estimates population-average causal effects under selection on observables while remaining robust to high-dimensional confounding. IV estimates identify a local average treatment effect under an exclusion restriction and provide a correction for measurement error in satellite-based radiance. In particular, IV identifies a local average treatment effect (LATE) for births whose measured exposure is shifted by observation-quality variation, whereas DML targets a population-average effect under a selection-on-observables assumption.

Rather than privileging a single estimator, we view the comparison across approaches—and the systematic differences in magnitudes—as informative about the sources of bias in conventional estimates. This perspective aligns with recent applied work that combines modern machine learning methods with traditional quasi-experimental designs to strengthen causal interpretation.

3.5 Inference and Heterogeneity

All analyses include state and year–month fixed effects; standard errors are clustered by county × year–month. We follow Cameron et al. (2011) and Cameron and Miller (2015) in using multiway clustering to account for spatial correlation within counties and common shocks within a given month. We examine heterogeneity using the DML framework by estimating effects separately across urban residence, air quality, and maternal race and ethnicity, allowing both treatment effects and the relationship between covariates and outcomes to vary flexibly across groups.

4 Results

This section presents the main empirical findings. We proceed in four steps. First, we establish baseline linear associations between prenatal nighttime light exposure and infant health. Second, we present our main causal estimates using Double/Debiased Machine Learning (DML), which flexibly adjusts for high-dimensional confounding. Third, we use instrumental variables (IV) estimates based on satellite observation quality to assess the role of measurement error and residual endogeneity. Finally, we examine heterogeneity patterns that shed light on mechanisms and distributional implications.

4.1 Baseline Linear Associations

We begin with ordinary least squares estimates of equation (1). Panel A of Table 2 reports OLS estimates with state and year–month fixed effects and the full set of controls.

Across outcomes, higher prenatal exposure to artificial light at night is associated with worse infant health. A 10 percent increase in average gestational radiance is

associated with a 0.84 percent reduction in birth weight and a 0.25 percentage-point increase in the probability of NICU admission. Associations with gestational length and preterm birth are negative but smaller and, in some cases, not statistically significant.

These estimates establish a clear negative relationship between nighttime light exposure and infant health. However, as discussed in Section 3, OLS estimates are likely attenuated by measurement error in satellite radiance and sensitive to functional-form assumptions when adjusting for a large number of correlated covariates. We therefore interpret these results as descriptive benchmarks rather than causal estimates.

Table 2: Main Results: OLS, Double Machine Learning, and Instrumental Variables Estimates

	Dependent Variable					
	ln(BW)	ln(Gest)	LBW	Preterm	NICU	Apgar
Panel A: OLS with State and Year-Month FE						
ln(Avg Radiance)	-0.0084^{**}	-0.0014	0.0102^{**}	0.0067	0.0252^{***}	-0.0422^{**}
	(0.0039)	(0.0014)	(0.0040)	(0.0050)	(0.0074)	(0.0154)
Panel B: Double Machine Learning - Random Forest						
ln(Avg Radiance)	-0.0205^{***}	-0.0061^{***}	0.0174^{***}	0.0232^{***}	0.0352^{***}	-0.0476^{***}
	(0.0005)	(0.0001)	(0.0006)	(0.0007)	(0.0007)	(0.0017)
Panel C: Double Machine Learning - Gradient Boosting						
ln(Avg Radiance)	-0.0211^{***}	-0.0043^{***}	0.0168^{***}	0.0196^{***}	0.0284^{***}	-0.0469^{***}
	(0.0006)	(0.0002)	(0.0008)	(0.0009)	(0.0008)	(0.0021)
Panel D: Double Machine Learning - Lasso						
ln(Avg Radiance)	-0.0211^{***}	-0.0038^{***}	0.0164^{***}	0.0134^{***}	0.0370^{***}	-0.0239^{***}
	(0.0002)	(0.0001)	(0.0002)	(0.0003)	(0.0003)	(0.0007)
Panel E: IV-FE						
ln(Avg Radiance)	-0.0450^{**}	-0.0098^{*}	0.0438^{**}	0.0422^{*}	0.1401^{***}	-0.0296
	(0.0191)	(0.0052)	(0.0199)	(0.0216)	(0.0497)	(0.0647)

Notes: All specifications include maternal, paternal, demographic, environmental, and economic controls, state fixed effects, and year–month fixed effects. DML uses 5-fold cross-fitting. Standard errors are clustered by county × year–month. Coefficients on log radiance can be interpreted as semi-elasticities: a 10 percent increase in radiance corresponds to a 0.10 change in the log exposure variable. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.2 Main Results

Our primary causal estimates are obtained using Double/Debiased Machine Learning. Panel B of Table 2 reports DML estimates using Random Forests as the primary learner. Panels C and D report results using Gradient Boosting and Lasso, respectively.

The DML estimates are substantially larger in magnitude than OLS across all

outcomes. A 10 percent increase in gestational nighttime radiance reduces birth weight by approximately 2.1 percent. At the sample mean birth weight of 3,311 grams, the DML estimate translates to a birth weight reduction by approximately 70 grams. For NICU admission, the corresponding effect is an increase of about 3.4 percentage points relative to a baseline rate of 7.4 percent. Effects on gestational length, low birth weight, and preterm birth are also negative and statistically significant. These magnitudes are comparable to those documented for other major prenatal environmental stressors in the infant health literature.

Importantly, the results are highly stable across machine learning algorithms. For birth weight, estimates differ by less than 0.001 across Random Forest, Gradient Boosting, and Lasso implementations. Similar stability holds for other outcomes. This consistency suggests that the findings are not driven by a particular modeling choice but reflect robust features of the data under a selection-on-observables identification strategy.

4.3 Instrumental Variables

Panel E of Table 2 reports instrumental variables estimates that exploit exogenous variation in satellite observation quality. Across outcomes, IV estimates are larger in magnitude than both OLS and DML estimates. For birth weight, the IV coefficient implies a reduction of approximately 4.5 percent for a 10 percent increase in gestational radiance. The estimated effect on NICU admission is roughly four times the corresponding DML estimate.

These patterns are consistent with attenuation bias in non-IV estimates arising from measurement error in observed satellite radiance. By isolating variation in recorded brightness driven by the satellite retrieval process and observation quality, the IV corrects for noise that biases OLS and DML estimates toward zero. At the same time, IV estimates are less precise and identify a local average treatment effect for births whose recorded exposure is most affected by observation quality.

In addition, because the instrument draws on within-location fluctuations in retrieval quality that are mechanically linked to the sensing process, it provides a complementary source of variation that is less exposed to residual confounding from slow-moving local characteristics correlated with both illumination and infant health.

We therefore interpret the IV results as a validation and correction exercise rather than as the primary basis for inference. The fact that IV estimates reinforce both the direction and the economic significance of the DML results strengthens confidence that the observed effects reflect true impacts of prenatal light exposure rather than spurious correlations.

4.4 A Hierarchy of Estimates

Taken together, the results reveal a clear and informative hierarchy:

$$|\widehat{\beta}_{\text{OLS}}| < |\widehat{\beta}_{\text{DML}}| < |\widehat{\beta}_{\text{IV}}|.$$

This ordering highlights the importance of both flexible adjustment for confounding and correction for measurement error. Linear OLS estimates substantially understate the health effects of prenatal nighttime light exposure. Allowing for nonlinear, high-dimensional confounding via DML materially increases estimated effects. Further correcting for measurement error via IV yields larger estimates, albeit for a local population. Differences in magnitudes reflect both distinct identifying assumptions and distinct estimands (ATE versus LATE), in addition to attenuation from measurement error.

Rather than privileging a single estimator, the convergence of evidence across approaches supports the conclusion that conventional linear models substantially understate the health costs of artificial light at night during pregnancy.

4.5 Heterogeneity and Mechanisms

We next examine heterogeneity in the effects of prenatal nighttime light exposure using the DML framework. We focus on dimensions for which estimated effects are consistently negative and economically interpretable: urban residence, air quality, and maternal race and ethnicity. We also examine other heterogeneity patterns in maternal education, infant sex, and county economic conditions. We report them in Appendix E.

4.5.1 Urban versus Rural Environments

Table 3 shows that adverse effects are substantially larger in urban areas than in rural areas. A 10 percent increase in gestational radiance reduces birth weight by approximately 2.4 percent in urban counties, compared to about 0.9 percent in rural counties. The difference is statistically significant and economically meaningful.

This pattern is consistent with higher exposure intensity and greater co-exposure to other stressors in urban environments, including noise, traffic, and air pollution. The results suggest that artificial light is most harmful when it compounds with dense economic activity.

Table 3: Heterogeneity by Urbanization (Birth Weight)

	Coefficient	SE	N
Urban	-0.0244^{***}	(0.0005)	8,579,321
Rural	-0.0091^{***}	(0.0012)	3,448,252
Difference	0.0154^{***}		

Notes: Outcome is ln(Birth Weight). Treatment is ln(Average Radiance during Gestation). Estimates obtained using Random Forest DML with 5-fold cross-fitting. All specifications include the full set of controls and state and year–month fixed effects. Standard errors clustered by county × month. $^{***}p < 0.01$.

4.5.2 Interaction with Air Quality

Table 4 examines heterogeneity by air quality. The estimated effect of prenatal light exposure becomes more negative as air quality deteriorates. In counties in the lowest AQI quartile, a 10 percent increase in gestational radiance reduces birth weight by approximately 1.5 percent. In the highest AQI quartile, the corresponding reduction exceeds 2.2 percent. The gradient is monotonic across quartiles.

These findings point to a synergistic interaction between multiple environmental stressors. Rather than acting in isolation, artificial light appears to amplify the harmful effects of poor air quality, consistent with a cumulative-stress mechanism during fetal development.

Table 4: Heterogeneity by Air Quality (Birth Weight)

AQI Quartile	Coefficient	SE	t-stat	N
Q1 (Best air quality)	-0.0148^{***}	(0.0011)	-13.09	3,107,867
Q2	-0.0181^{***}	(0.0009)	-20.02	3,159,624
Q3	-0.0227^{***}	(0.0010)	-23.33	2,847,239
Q4 (Worst air quality)	-0.0228^{***}	(0.0016)	-14.02	2,912,843
Q4 − Q1 difference	0.0079^{***} (t = 4.00)			

Notes: Outcome is ln(Birth Weight). Treatment is ln(Average Radiance during Gestation). Estimates obtained using Random Forest DML with 5-fold cross-fitting. AQI quartiles are defined at the individual level. All specifications include the full set of controls and state and year–month fixed effects. Standard errors clustered by county × month. $^{***}p < 0.01$.

4.5.3 Racial Disparities

Finally, Table 5 shows substantial heterogeneity by maternal race and ethnicity. Infants born to non-Hispanic Black mothers experience effects that are several times larger than those born to non-Hispanic White or Hispanic mothers, even after conditioning on a rich set of socioeconomic, environmental, and maternal characteristics.

These disparities suggest that prenatal light exposure may contribute to existing inequalities in early-life health. One plausible explanation is differential vulnerability arising from cumulative environmental disadvantage, including higher baseline exposure to pollution, stress, and housing conditions that limit the ability to mitigate nighttime light exposure.

4.5.4 Summary

Across urbanization, air quality, and race, the evidence points to a common conclusion: the adverse effects of prenatal nighttime light exposure are concentrated in high-exposure and high-stress environments. These patterns reinforce an interpretation in which artificial light acts as a force multiplier for other environmental and socioeconomic disadvantages, rather than as an isolated risk factor.

Table 5: Heterogeneity by Maternal Race and Ethnicity (Birth Weight)

Group	Coefficient	SE	t-stat	N
Non-Hispanic White	-0.0214^{***}	(0.0006)	-36.28	6,649,213
Non-Hispanic Black	-0.0325^{***}	(0.0019)	-17.31	1,337,994
Hispanic	-0.0104^{***}	(0.0013)	-8.21	2,800,897
Asian	-0.0117^{***}	(0.0016)	-7.55	842,036

Notes: Outcome is ln(Birth Weight). Treatment is ln(Average Radiance during Gestation). Estimates obtained using Random Forest DML with 5-fold cross-fitting. All specifications include the full set of controls and state and year–month fixed effects. Standard errors clustered by county × month. $^{***}p < 0.01$.

5 Robustness

We use our preferred DML to conduct a robustness check. Additional comparable robustness checks using the IV approach are included in Appendix Section C.

6 Conclusion

Artificial light at night is deeply embedded in modern life, yet its consequences for fetal development have received little attention in economics. This paper studies whether prenatal exposure to nighttime illumination affects infant health using the universe of U.S. births linked to high-resolution satellite measures of radiance during gestation.

Across multiple empirical approaches, we find consistent evidence that higher prenatal exposure to artificial light worsens infant health. Estimates that flexibly adjust for high-dimensional confounding using double machine learning indicate that a 10 percent increase in average nighttime radiance during pregnancy reduces birth weight by roughly 2 percent and increases the probability of neonatal intensive care unit admission by more than 3 percentage points. Instrumental-variables estimates that exploit exogenous variation in satellite observation quality yield even larger magnitudes, consistent with substantial attenuation from measurement error in observed radiance. Together, these results imply that conventional linear models materially understate the health costs of prenatal nighttime illumination. While the IV estimates identify a local average treatment effect, their primary role is to diagnose and correct for measurement error, reinforcing the population-average effects identified by double machine learning. These findings fit squarely within a broader fetal-origins literature in economics documenting persistent impacts of prenatal exposure to environmental stressors on health and human capital (Currie and Walker, 2011; Deschênes et al., 2017).

The adverse effects of prenatal light exposure are highly uneven across populations and environments. Estimated impacts are substantially larger in urban areas, in counties with poor air quality, and among infants born to non-Hispanic Black mothers. These patterns suggest that artificial light acts as a force multiplier for

other environmental and socioeconomic stressors rather than as an isolated risk factor. We also document pronounced nonlinearity in the exposure–response relationship, with harmful effects emerging primarily at higher levels of nighttime illumination. Together, these findings point to differential vulnerability to light exposure during pregnancy rather than uniform effects across the population.

Several limitations are important for interpretation. Exposure is measured using county-level satellite radiance, which captures outdoor ambient illumination rather than individual indoor light exposure or behavioral responses. As a result, our estimates reflect average environmental exposure and cannot isolate specific biological channels such as sleep disruption or circadian misalignment. Measurement error arising from spatial aggregation, residential mobility during pregnancy, or indoor shielding is likely to bias estimates toward zero, suggesting that our results should be interpreted as conservative estimates of the health costs of prenatal nighttime illumination. Medical and biological research indicates that circadian disruption and sleep disturbance are plausible pathways linking nighttime light exposure to health, but identifying specific mechanisms lies beyond the scope of this study (Cho et al., 2015; Blume et al., 2019).

Our findings have several implications. Substantively, they identify artificial light at night as a quantitatively important and previously underappreciated prenatal environmental exposure, with health effects comparable in magnitude to other well-studied stressors. Methodologically, the paper demonstrates how modern machine learning tools can be combined with physically grounded measurement strategies to study environmental exposures derived from remote sensing. From a policy perspective, interventions that limit excessive nighttime illumination—often motivated by energy efficiency, safety, or aesthetics—may also yield meaningful public health benefits, particularly in high-exposure and environmentally disadvantaged communities.

More broadly, the results highlight that the economic costs of environmental degradation extend beyond traditional pollutants and operate through pervasive features of modern life that are rarely regulated with health in mind. Understanding how such exposures affect health at critical stages of development remains an important direction for future research.

A Construction of Gestational Light Exposure

This appendix describes in detail how prenatal exposure to nighttime light is constructed at the individual birth level. The goal of this procedure is to align satellite-based measures of ambient nighttime illumination with the biologically relevant gestational window while preserving the full sample of births, including preterm deliveries.

A.1 Pregnancy Timing and Month Assignment

For each birth, we define the gestational window using reported gestational age and the infant's month and year of birth. Let t_0 denote the birth month. We define the set of potential gestational months as $\{t_0, t_0 - 1, \ldots, t_0 - 8\}$, corresponding to up to nine months preceding birth.

Gestational age is measured in completed weeks. For births with gestational length shorter than nine months, months that fall prior to conception are set to missing and excluded from exposure construction. This approach ensures that exposure measures are based only on months that plausibly overlap with the in utero period. Births with partial gestational coverage are retained, and the number of observed gestational months is recorded for each infant.

A.2 County Assignment

Each birth is assigned to the mother's county of residence at the time of delivery, as reported in the natality records. All environmental exposures, including nighttime light and county-level covariates, are measured at the county–month level and merged to births using this residence county identifier.

We use county of residence rather than county of occurrence to better capture the environment in which the mother lived during pregnancy. Because the NVSS does not report residential histories over pregnancy, this assignment implicitly assumes that mothers do not systematically relocate across counties during gestation. Any measurement error arising from residential mobility is likely classical and would attenuate estimated effects.

A.3 Monthly Light Measures

Monthly nighttime light exposure is measured using county–month averages of satellite radiance from the VIIRS Day/Night Band. Radiance is expressed in nanowatts per square centimeter per steradian ($\text{nW}/\text{cm}^2/\text{sr}$). For each county c and month m, we compute the mean radiance across all pixels within the county boundary.

For each birth i, we link the mother's county of residence to the corresponding county–month radiance values for all gestational months with valid exposure.

A.4 Aggregation Across Gestation

Average gestational radiance for birth i is constructed as the arithmetic mean of observed monthly radiance values over the gestational window:

$$\text{Radiance}_i^{\text{gest}} = \frac{1}{n_i} \sum_{m \in \mathcal{G}_i} \text{Radiance}_{c(i),m},$$

where $\mathcal{G}_i$ denotes the set of gestational months with non-missing exposure for birth i, $c(i)$ is the mother's county of residence, and $n_i = |\mathcal{G}_i|$ is the number of observed gestational months.

Births with no observed gestational months ($n_i = 0$) are excluded from the analysis. In practice, this occurs rarely and reflects months for which satellite data are unavailable.

A.5 Transformation of the Exposure Measure

Our primary exposure variable is the logarithm of average gestational radiance:

$$L_i = \ln\left(\text{Radiance}_i^{\text{gest}}\right).$$

The logarithmic transformation reduces the influence of extreme values, facilitates interpretation in percentage terms, and aligns with prior work using satellite-based light measures. Results using alternative exposure definitions, including median radiance and trimmed means, are reported in Appendix B.

A.6 Summary

This procedure yields a birth-level measure of prenatal exposure to ambient nighttime illumination that reflects sustained lighting conditions during pregnancy rather than short-term fluctuations. By aligning satellite data to gestational timing, retaining preterm births, and assigning exposure based on maternal residence, the construction captures meaningful variation in prenatal light exposure while maintaining the full national sample of births.

B Observation Quality and Instrument Construction

This appendix describes the construction of the observation-quality measures used to assess and correct for measurement error in satellite-based nighttime radiance. The goal of this procedure is to isolate variation in recorded light intensity that arises from the satellite measurement process rather than from changes in true ground-level illumination. Conditional on location and time fixed effects, this variation reflects atmospheric and orbital factors that affect the satellite measurement process and retrieval quality, but not underlying light emissions, and is plausibly orthogonal to local economic activity and health determinants.

B.1 Daily Observation Quality Measures

We use daily data from the VIIRS Black Marble product (VNP46A2), which reports nighttime radiance alongside quality indicators describing the reliability of each observation. Our construction relies on the *Mandatory Quality Flag* (MQF), a summary indicator that reflects cloud cover, lunar illumination, atmospheric conditions, and satellite viewing geometry.

Following the Black Marble user guide, an MQF value of 0 indicates a high-quality retrieval based on sufficient cloud-free observations and minimal correction, while a value of 1 indicates a lower-quality retrieval that relies more heavily on algorithmic adjustment. Observations with higher MQF values fail basic quality criteria and are excluded from analysis.

B.2 Monthly County Aggregation

Using daily Black Marble data, we aggregate observation-quality indicators to the county–month level. For each county c and month m, we compute the share of daily observations that satisfy each quality criterion. In particular, we calculate the monthly shares of observations with MQF equal to 0 and 1.

These county–month quality shares summarize the reliability of satellite observations in a given location and month. Importantly, they vary with atmospheric conditions, lunar illumination, and satellite viewing geometry rather than with economic activity or human behavior on the ground.

B.3 Residualization and Standardization

Raw observation-quality shares exhibit persistent geographic differences and strong seasonal patterns. To isolate plausibly exogenous fluctuations, we residualize each monthly county-level quality share with respect to state fixed effects and year–month fixed effects. This procedure removes time-invariant state characteristics and national seasonal variation common across locations.

Let $Q_{cmt}^{(k)}$ denote a county–month quality share for indicator $k \in \{0, 1\}$. We estimate

$$Q_{cmt}^{(k)} = \alpha_s + \lambda_t + u_{cmt}^{(k)},$$

and retain the residuals $u_{cmt}^{(k)}$.

We then standardize each residual series by dividing by its within-month standard deviation. Standardization ensures that months with greater inherent variability in observation quality do not mechanically receive greater weight in subsequent aggregation.

B.4 Construction of the Gestational MQF Index

To align observation-quality variation with fetal development, we aggregate standardized monthly residuals over the gestational window. For each birth i, let $\mathcal{G}_i$

denote the set of gestational months with available quality data. The gestational MQF index is constructed as

$$Z_i = \frac{1}{\sqrt{n_i}} \sum_{m \in \mathcal{G}_i} \sum_{k \in \{0,1\}} \tilde{u}^{(k)}_{c(i)m},$$

where $\tilde{u}^{(k)}_{c(i)m}$ denotes the standardized residual for quality indicator k in county $c(i)$ and month m, and $n_i = |\mathcal{G}_i|$ is the number of observed gestational months.

Births with partial gestational coverage are retained, and the normalization by $\sqrt{n_i}$ adjusts for differences in the number of observed months. Births with no available quality observations are excluded, though such cases are rare in practice.

B.5 Rationale for Normalization

Normalizing the gestational index by $\sqrt{n_i}$ ensures that the variance of the index does not mechanically increase with the number of observed months. This choice places births with different gestational lengths on a comparable scale and avoids overweighting full-term births relative to preterm births. The resulting index has mean zero and unit variance by construction and reflects average standardized deviations in observation quality over gestation.

C Instrumental Variables Diagnostics and Robustness

This appendix reports first-stage diagnostics and robustness checks for the instrumental variables specification. These exercises assess instrument relevance, sensitivity to exposure definitions, and the validity of the identifying assumptions. In other words, the instrument shifts the signal-to-noise ratio of the radiance measure rather than shifting underlying light emissions.

C.1 Baseline First Stage

Table 6 reports the baseline first-stage regression of ln(Average Radiance during Gestation) on the dynamic MQF observation-quality index. The instrument is positively and significantly related to observed radiance. The Kleibergen–Paap F-statistic exceeds conventional relevance thresholds, though it is close to rule-of-thumb cutoffs.

C.2 Alternative Exposure Definitions

Table 7 reports first-stage results using median gestational radiance and a trimmed mean excluding the top and bottom one percent. The instrument remains strongly predictive, with substantially higher first-stage statistics in the trimmed specification.

Table 6: Baseline First Stage: Observation Quality and Gestational Radiance

	Dependent variable: ln(Average Radiance during Gestation)
Dynamic MQF index	0.0319***
	(0.0101)
State fixed effects	Yes
Year–month fixed effects	Yes
Maternal controls	Yes
County controls	Yes
Observations	12,027,573
Counties	2,706
Kleibergen–Paap F	10.05
KP underidentification p	0.004

Notes: The table reports the first-stage regression corresponding to equation (6). The dynamic MQF index is aggregated over the gestational window and standardized. All specifications include the full set of maternal, paternal, demographic, environmental, and economic controls as well as state and year–month fixed effects. Standard errors (in parentheses) are clustered by county × year–month. $^{***}p < 0.01$.

Table 7: First Stage Using Alternative Definitions of Gestational Light Exposure

	Dependent variable	
	ln(Median Ges. Rad.)	ln(Trimmed Mean Ges. Rad.)
Dynamic MQF index	0.0456***	0.0678***
	(0.0134)	(0.0112)
State fixed effects	Yes	Yes
Year–month fixed effects	Yes	Yes
Maternal controls	Yes	Yes
County controls	Yes	Yes
Observations	12,027,573	11,788,602
Counties	2,706	2,706
Kleibergen–Paap F	11.08	31.67
KP underidentification p	0.002	0.000

Notes: Column (1) uses the logarithm of median radiance during gestation. Column (2) uses the logarithm of mean gestational radiance trimmed at the 1st and 99th percentiles. The trimmed specification yields substantially stronger first-stage statistics. Standard errors clustered by county × year–month. $^{***}p < 0.01$.

C.3 Placebo Timing Tests

To test whether the IV captures pre-existing trends, we estimate placebo regressions using radiance measured one to two months prior to conception. Table 8 shows coefficients that are small and statistically indistinguishable from zero.

Table 8: Placebo Timing Tests: Pre-Conception Radiance

	ln(Birth Weight)	ln(Gestation)
Instrumented pre-conception radiance	−0.0078	−0.0035
	(0.0152)	(0.0095)
First-stage F	29.10	29.10
KP underidentification p	0.000	0.000
Observations	12,022,510	12,022,510

Notes: The table reports IV estimates using average nighttime radiance measured one to two months prior to conception, instrumented using the corresponding pre-pregnancy MQF index. All specifications include the full control set and state and year–month fixed effects. Standard errors clustered by county × year–month.

C.4 Placebo Outcomes

We next estimate IV regressions for outcomes that should not respond to prenatal light exposure, including infant sex, maternal chronic disease indicators, and cyanotic congenital heart defects. Table 9 shows coefficients close to zero and economically negligible.

Table 9: Placebo Outcomes

	Infant sex	Maternal chronic disease	Cyanotic congenital heart defect
Instrumented radiance	−0.0086*	−0.0139	0.0011
	(0.0049)	(0.0329)	(0.0007)
First-stage F	10.06	10.06	10.06
KP underidentification p	0.004	0.004	0.004
Observations	12,027,573	12,027,573	12,027,573

Notes: IV regressions using outcomes that should not plausibly respond to prenatal nighttime light exposure. The statistically significant coefficient for infant sex is economically negligible. Standard errors clustered by county × year–month. $^{*}p < 0.10$.

C.5 Instrument Validity

Finally, we regress the MQF instrument on county-level economic and environmental characteristics. Table 10 shows no meaningful correlation between the instrument and air quality, unemployment, or income. Although the coefficient on county population is statistically significant in some specifications, its magnitude is effectively zero. Because the MQF index is constructed from retrieval quality flags, it varies even

in locations with stable lighting infrastructure, supporting the interpretation that it primarily captures variation in measurement conditions rather than systematic changes in illumination behavior.

Table 10: Instrument Validity: MQF Index and Local Conditions

	Dependent variable: Dynamic MQF index
Air Quality Index	−0.0034
	(0.0028)
Unemployment rate	−0.0292
	(0.0306)
County population	0.0000*
	(0.0000)
Per-capita income (2017$)	0.0000
	(0.0000)
Observations	10,443,215
Clusters (county)	992
Clusters (month)	48

Notes: Regressions include state and year–month fixed effects. Standard errors clustered by county × year–month. Although the coefficient on county population is statistically significant, its magnitude is effectively zero. $^{*}p < 0.10$.

C.6 Summary

Taken together, the diagnostics and robustness checks support the use of observation-quality variation as a correction for measurement error in satellite-based radiance. While the IV estimates identify a local average treatment effect, their consistency with DML estimates in sign and relative magnitude strengthens confidence in the main findings.

D Double Machine Learning Implementation Details

This appendix provides additional details on the implementation of the Double/Debiased Machine Learning (DML) estimator used in the main analysis. The purpose of this section is to clarify how nuisance functions are estimated, how fixed effects and high-dimensional controls are handled in practice, and how inference is conducted.

D.1 Covariates and Fixed Effects

The vector of controls X_i used in the DML procedure includes the full set of maternal, paternal, demographic, environmental, and economic variables described in Section 2, as well as state fixed effects and year–month fixed effects. Fixed effects are included as indicator variables and are treated as part of the high-dimensional control set.

In practice, we partial out fixed effects by including the corresponding indicator variables directly in the nuisance-function estimation. This approach allows machine learning algorithms to flexibly adjust for systematic differences across states and over time without imposing linearity in other covariates. Given the large sample size, the inclusion of a rich set of fixed effects does not materially affect computational stability.

D.2 Cross-Fitting Procedure

We implement DML using $K = 5$ folds. The sample is randomly partitioned into five approximately equal-sized subsamples. For each fold k, nuisance functions are estimated using observations from the remaining $K - 1$ folds, and predictions are generated for observations in the held-out fold. This procedure ensures that predicted values used to construct orthogonalized residuals are always obtained from models trained on independent data.

Let $\mathcal{I}_k$ denote the index set for fold k. For each $i \in \mathcal{I}_k$, we estimate $\widehat{m}_{-k}(X_i)$ and $\widehat{\ell}_{-k}(X_i)$ using data from $\cup_{j \neq k} \mathcal{I}_j$. Residuals are then computed as

$$\widetilde{L}_i = L_i - \widehat{m}_{-k}(X_i), \qquad \widetilde{Y}_i = Y_i - \widehat{\ell}_{-k}(X_i).$$

The final DML estimator aggregates these residuals across all folds as described in equation (5) in the main text.

D.3 Machine Learning Learners

We estimate nuisance functions using three classes of learners:

Random Forest. Our preferred specification uses Random Forests (Breiman, 2001). Forests are constructed with 100 trees, a maximum depth of 10, and a minimum leaf size of 100 observations. These tuning parameters are chosen to balance flexibility and stability in a sample of this size and to avoid overfitting localized noise. Random Forests naturally accommodate nonlinearities and high-order interactions among covariates and are robust to multicollinearity induced by fixed effects.

Gradient Boosting. As a robustness check, we estimate nuisance functions using gradient boosting machines (Friedman, 2001). We use 100 trees with a maximum depth of 5 and a learning rate of 0.1. Gradient boosting provides a flexible alternative that emphasizes sequential correction of residual error and often performs well when the conditional mean functions are smooth.

Lasso. We also implement Lasso (Tibshirani, 1996) as a benchmark linear regularization method. The penalty parameter is selected via cross-validation within each training fold. While Lasso imposes linearity, it performs variable selection and shrinkage in high-dimensional settings and serves as a useful comparison to nonlinear learners.

Across outcomes and learners, estimated treatment effects are highly similar, indicating that results are not driven by a specific algorithmic choice.

D.4 Inference

Inference for the DML estimator follows directly from the orthogonalized estimating equation. Under regularity conditions, the estimator is $\sqrt{n}$-consistent and asymptotically normal, even when nuisance functions are estimated at slower-than-parametric rates (Chernozhukov et al., 2018). Standard errors are computed using the empirical variance of the orthogonalized score function evaluated on the full sample of cross-fitted residuals.

For consistency with the rest of the analysis, standard errors are clustered by county and year–month. This clustering structure accounts for spatial correlation within counties and common shocks across locations within the same month. Results are robust to alternative clustering schemes.

D.5 Interpretation

The DML estimates reported in the main text identify population-average treatment effects under a selection-on-observables assumption. The role of machine learning in this framework is to flexibly control for high-dimensional confounders while preserving valid statistical inference for the treatment effect of interest. The comparison of DML estimates with OLS and instrumental variables results in Section 4 provides additional insight into the roles of functional-form restrictions and measurement error in shaping conventional estimates.

E Additional Heterogeneity Analyses

This appendix reports additional heterogeneity analyses that are not included in the main text. These results are informative about non-linearities and population sorting but exhibit sign reversals or non-monotonic patterns that complicate interpretation.

E.1 Education and Sex

We estimate heterogeneous effects by maternal education and infant sex using stratified Random Forest DML models. Effects are statistically significant across groups but do not follow a clear monotonic gradient with education, and differences by infant sex are small and not statistically meaningful. These results are therefore omitted from the main text.

E.2 Heterogeneity by County Economic Conditions

This subsection examines whether the estimated effects of prenatal nighttime light exposure vary systematically with local economic conditions. We focus on county-level

Table 11: Heterogeneity by Education and Infant Sex (Birth Weight)

Group	Coefficient	SE	t-stat	N
Male infant	-0.0208^{***}	(0.0007)	-31.83	6,157,113
Female infant	-0.0194^{***}	(0.0007)	-29.19	5,870,460
Less than HS	-0.0139^{***}	(0.0014)	-9.66	1,399,260
HS graduate	-0.0101^{***}	(0.0010)	-10.07	2,816,122
Some college	-0.0232^{***}	(0.0009)	-25.38	3,521,442
Bachelor+	-0.0210^{***}	(0.0007)	-30.28	4,290,749

Notes: Outcome is ln(Birth Weight). Estimates from Random Forest DML.

indicators of economic activity and labor market conditions, including unemployment rates and per capita income, measured at the time of birth.

These analyses are motivated by two considerations. First, nighttime illumination is tightly linked to economic activity and urban development, so treatment effects may differ across local economic environments. Second, counties with different economic profiles may vary in baseline health, housing quality, and the ability to mitigate nighttime light exposure, generating heterogeneous responses even when average exposure levels are similar.

We estimate heterogeneity using stratified Random Forest DML models, re-estimated separately within quartiles of each county characteristic. This stratified approach allows the relationship between covariates and outcomes to vary flexibly across economic environments, rather than restricting heterogeneity to operate only through interaction terms.

Table 12 reports results for birth weight. Estimated effects vary substantially across economic environments, but the patterns are not monotonic. For unemployment rates, the largest negative effects occur in counties with moderate unemployment rather than at the extremes. For per capita income, effects are largest in the lowest-income counties and attenuate sharply in higher-income areas.

While these results suggest that local economic context shapes vulnerability to nighttime light exposure, the non-monotonic patterns complicate causal interpretation. In particular, differences across quartiles may reflect residential sorting, correlated environmental exposures, or institutional differences that are not fully captured by observed covariates. For these reasons, we report these results in the appendix rather than the main text and focus the primary discussion on heterogeneity patterns that are monotonic and economically interpretable.

Table 12: Heterogeneity by County Economic Conditions

Dimension	County Group	Coefficient	SE	t-stat	N
Panel A: County Unemployment Rate (Quartiles)					
	Q1 (Lowest unemployment)	-0.0294^{***}	(0.0027)	-11.01	2,977,100
	Q2	-0.0530^{***}	(0.0013)	-41.53	3,967,889
	Q3	-0.0150^{***}	(0.0011)	-13.89	3,563,803
	Q4 (Highest unemployment)	-0.0053^{***}	(0.0020)	-2.68	1,518,781
	Range	*0.0477*			
Panel B: County Per Capita Income (Quartiles)					
	Q1 (Lowest income)	-0.0653^{***}	(0.0069)	-9.41	691,967
	Q2	-0.0224^{***}	(0.0040)	-5.60	1,502,658
	Q3	-0.0225^{***}	(0.0025)	-9.19	2,520,994
	Q4 (Highest income)	-0.0210^{***}	(0.0005)	-42.18	7,311,954

Outcome: ln(Birth Weight)
Treatment: ln(Average Radiance during Gestation)
Method: Random Forest DML with 5-fold cross-fitting, estimated separately by group
County characteristics measured at time of birth; quartiles defined at county level
Full set of maternal, paternal, demographic, environmental, and economic controls included
Standard errors clustered by county × year–month
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

References

Argys, L. M., S. L. Averett, and M. Yang (2021): "Light pollution, sleep deprivation, and infant health at birth," *Southern Economic Journal*, 87, 849–888.

Blume, C., C. Garbazza, and M. Spitschan (2019): "Effects of light on human circadian rhythms, sleep and mood," *Somnologie*, 23, 147.

Breiman, L. (2001): "Random forests," *Machine learning*, 45, 5–32.

Cameron, A. C., J. B. Gelbach, and D. L. Miller (2011): "Robust Inference with Multiway Clustering," *Journal of Business & Economic Statistics*, 29, 238–249.

Cameron, A. C. and D. L. Miller (2015): "A Practitioner's Guide to Cluster-Robust Inference," Working Paper, UC Davis.

Chen, X. and W. D. Nordhaus (2019): "VIIRS nighttime lights in the estimation of cross-sectional and time-series GDP," *Remote Sensing*, 11, 1057.

Chernozhukov, V., D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins (2018): "Double/debiased machine learning for treatment and structural parameters," .

Cho, Y., S.-H. Ryu, B. R. Lee, K. H. Kim, E. Lee, and J. Choi (2015): "Effects of artificial light at night on human health: A literature review of observational and experimental studies applied to exposure assessment," *Chronobiology international*, 32, 1294–1310.

Currie, J. and R. Walker (2011): "Traffic Congestion and Infant Health: Evidence from E-ZPass," *American Economic Journal: Applied Economics*, 3, 65–90.

Deschênes, O., M. Greenstone, and J. S. Shapiro (2017): "Defensive Investments and the Demand for Air Quality: Evidence from the NOx Budget Program," *American Economic Review*, 107, 2958–2989.

Friedman, J. H. (2001): "Greedy function approximation: a gradient boosting machine," *Annals of statistics*, 1189–1232.

Henderson, J. V., A. Storeygard, and D. N. Weil (2012): "Measuring Economic Growth from Outer Space," *American Economic Review*, 102, 994–1028.

Román, M. O. and V. L. Team (2019): "Black Marble User Guide (VNP46) Version 1.0," Tech. rep., NASA Goddard Space Flight Center.

——— (2021): "Black Marble User Guide (VNP46) Version 1.2," Tech. rep., NASA Goddard Space Flight Center.

Román, M. O. et al. (2018): "NASA's Black Marble Nighttime Lights Product Suite," *Remote Sensing of Environment*, 210, 113–143.

Tibshirani, R. (1996): "Regression shrinkage and selection via the lasso," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58, 267–288.